



2011 Half-Yearly Examination

FORM VI

MATHEMATICS EXTENSION 1

Thursday 3rd March 2011

General Instructions

- Writing time — 2 Hours
- Write using black or blue pen.
- Board-approved calculators and templates may be used.
- A list of standard integrals is provided at the end of the examination paper.
- All necessary working should be shown in every question.
- Start each question in a new booklet.

Structure of the paper

- Total marks — 84
- All seven questions may be attempted.
- All seven questions are of equal value.

Collection

- Write your candidate number clearly on each booklet.
- Hand in the seven questions in a single well-ordered pile.
- Hand in a booklet for each question, even if it has not been attempted.
- If you use a second booklet for a question, place it inside the first.
- Place the question paper inside your answer booklet for Question One.

Checklist

- SGS booklets — 7 per boy
- Candidature — 128 boys

Examiner

LYL

QUESTION ONE (12 marks) Use a separate writing booklet.

Marks

(a) Write down the exact value of:

(i) $\cos \frac{7\pi}{6}$

1

(ii) $\tan^{-1}(\sqrt{3})$

1

(b) Simplify $2 \sin 2x \cos 2x$.

1

(c) Differentiate with respect to x :

(i) $x^2 \cos x$

2

(ii) e^{4x}

1

(iii) $\ln(5x + 3)$

1

(d) Evaluate $\lim_{x \rightarrow 0} \left(\frac{\sin 7x}{x} \right)$. You must show working.

1

(e) Find:

(i) $\int (1 - x)^5 dx$

1

(ii) $\int (\cos x - \sin x) dx$

1

(iii) $\int \sin 2x dx$

1

(iv) $\int e^{2-3x} dx$

1

QUESTION TWO (12 marks) Use a separate writing booklet.

Marks

- (a) What is the amplitude and period of the function $f(x) = 3 \sin 2x$? 2
- (b) Find the acute angle between the lines $2x + y - 2 = 0$ and $y = x + 9$. Give your answer to the nearest degree. 2
- (c) Solve $\frac{1}{x+5} \geq 1$. 3
- (d) Consider the function $f(x) = x^2 - 9$, for $x \geq 0$.
- (i) Sketch and clearly label $y = f(x)$, for $x \geq 0$. 1
- (ii) Find the equation of the inverse function. 2
- (iii) State the domain of $y = f^{-1}(x)$. 1
- (iv) Sketch the inverse function $y = f^{-1}(x)$. 1

QUESTION THREE (12 marks) Use a separate writing booklet.

Marks

- (a) The point A is $(12, -10)$ and the point B is $(-3, -5)$. The point P divides the interval AB internally in the ratio $2:3$. Find the coordinates of P . 2
- (b) Consider the parabola with parametric equations
- $$x = 4t,$$
- $$y = t^2 - 1.$$
- (i) Find the Cartesian equation of this parabola. 2
- (ii) State the coordinates of the vertex. 1
- (iii) State the coordinates of the focus. 2
- (c) Solve $\cos 2\theta - \sin \theta = 0$, for $0 \leq \theta \leq 2\pi$. 3
- (d) Find the exact value of $\tan \left(\cos^{-1} \left(-\frac{5}{7} \right) \right)$. 2

QUESTION FOUR (12 marks) Use a separate writing booklet.

Marks

- (a) Prove the identity

2

$$\frac{1}{1 - \cos \theta} + \frac{1}{1 + \cos \theta} = 2 \operatorname{cosec}^2 \theta .$$

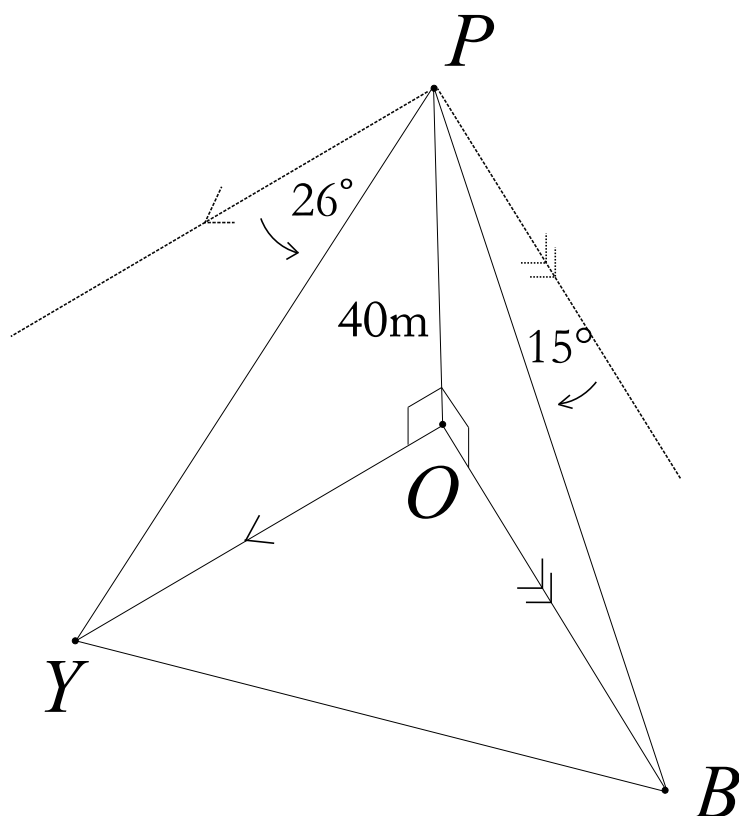
- (b) Evaluate $\int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \tan^2 x \, dx$.

3

- (c) A computer animation shows the sides of a cube increasing at a rate of 3 mm/s. Find the rate at which the volume V is increasing when the cube has a side length of 5 mm.

3

- (d)



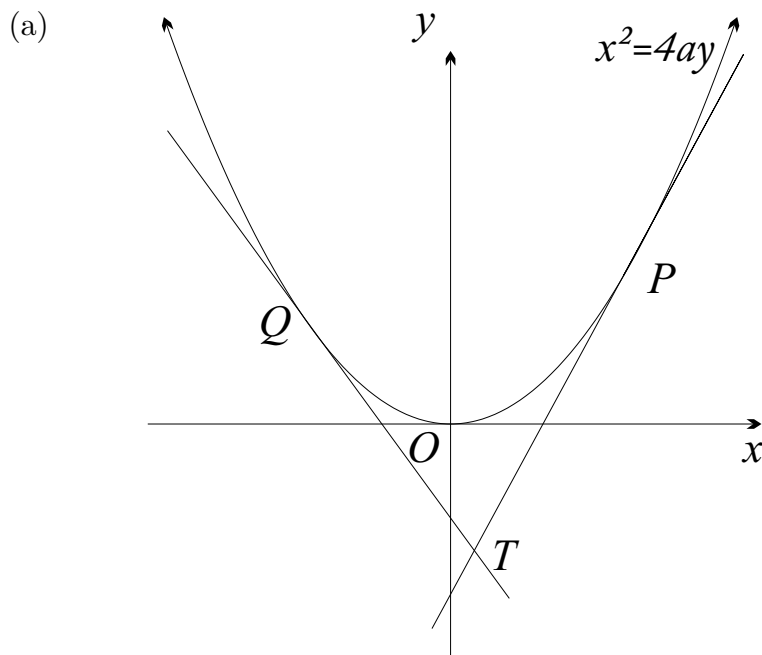
The diagram above shows an offshore oil rig PO . A viewing platform at P is 40 metres above its base O which is at sea level. A boat B and a yacht Y are observed from P with angles of depression of 15° and 26° respectively. From O , boat B is on a bearing of 150° and yacht Y is on a bearing of 225° .

- (i) Explain why $\angle YOB = 75^\circ$.

1

- (ii) Find the distance between the vessels, YB , to the nearest metre.

3

QUESTION FIVE (12 marks) Use a separate writing booklet.**Marks**

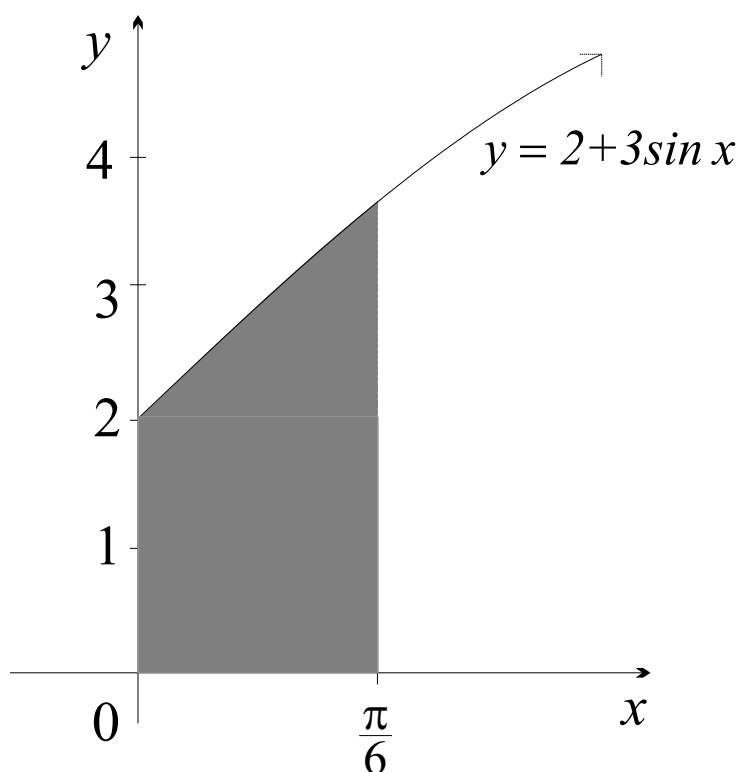
In the diagram above, two points $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ lie on the parabola $x^2 = 4ay$. The tangents at P and Q intersect at the point T .

The equation of the tangent at P is $y = px - ap^2$.

- (i) Show that the coordinates of the point T are $(a(p + q), apq)$. **3**
- (ii) Given that $pq = -2$, find the equation of the locus of T . **1**
- (b) (i) Express $\sqrt{3}\cos x - \sin x$ in the form $R\cos(x + \alpha)$, where $R > 0$ and $0 \leq \alpha \leq \frac{\pi}{2}$. **2**
- (ii) Hence, or otherwise, solve the equation $\sqrt{3}\cos x - \sin x = 1$, for $-\pi \leq x \leq \pi$. **2**
Give your solutions as exact values.
- (c) Use mathematical induction to prove that $7^{2n+1} + 11^{2n+1}$ is divisible by 3, **4**
for $n = 0, 1, 2, 3, \dots$

QUESTION SIX (12 marks) Use a separate writing booklet.**Marks**

(a)



The diagram above shows the region bounded by the curve $y = 2 + 3\sin x$ and the x -axis from $x = 0$ to $x = \frac{\pi}{6}$. Find the exact volume of the solid generated when the region is rotated about the x -axis.

3

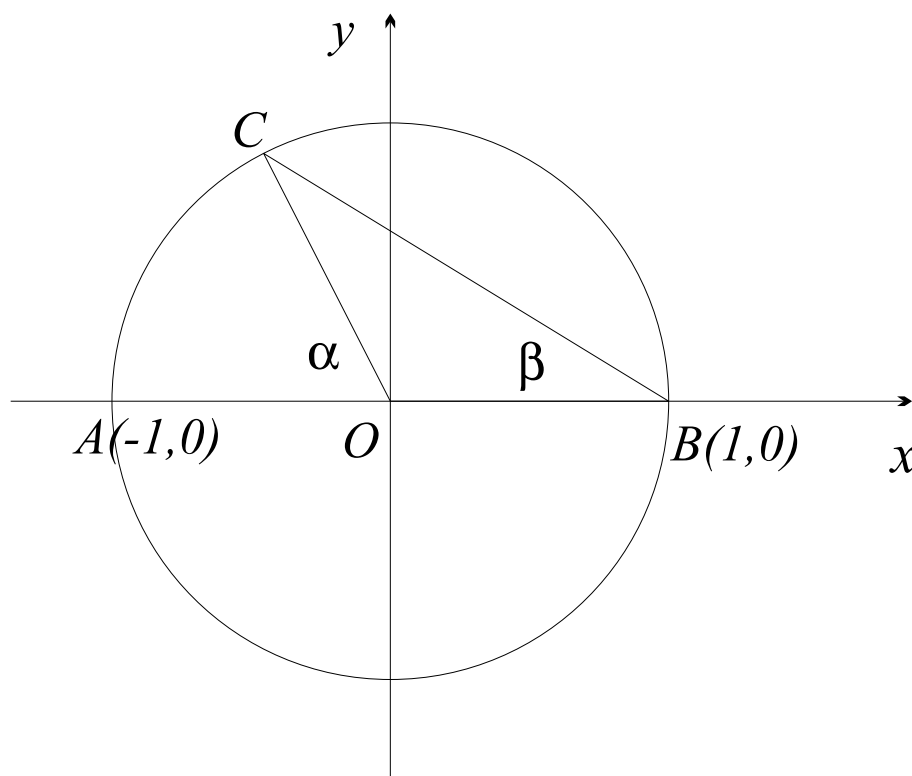
(b) Consider the curve whose equation is $f(x) = \frac{x(x-1)}{(x+1)(x-2)}$.

- (i) Find the coordinates of any intercepts with the axes.
- (ii) Find the equations of the vertical asymptotes of the curve.
- (iii) Find the equation of the horizontal asymptote of the curve.
- (iv) Find $f'(x)$.
- (v) Find the stationary point and determine its nature.
- (vi) Sketch the curve.

2**1****1****2****2****1**

QUESTION SEVEN (12 marks) Use a separate writing booklet.**Marks**(a) Consider the functions $y = 2 \sin x$ and $y = \tan x$, for $-\pi \leq x \leq \pi$.(i) On the same axes, sketch the graphs of the functions $y = 2 \sin x$ and $y = \tan x$, for $-\pi \leq x \leq \pi$. 2(ii) Find all the solutions of $2 \sin x = \tan x$, for $-\frac{\pi}{2} < x < \frac{\pi}{2}$. 2(iii) Hence solve $2 \sin x \leq \tan x$, for $-\frac{\pi}{2} < x < \frac{\pi}{2}$. 1

(b)



In the diagram above, the points $A(-1, 0)$, $B(1, 0)$, and C all lie on the circle with centre O and radius 1. Let $\angle COA = \alpha$ and $\angle CBO = \beta$.

(i) Given that the line BC has gradient m , find its equation. 1(ii) Show that the x -coordinates of B and C are solutions of the equation 2

$$(1 + m^2)x^2 - 2m^2x + m^2 - 1 = 0.$$

(iii) Using this equation, find the coordinates of C in terms of m . 2(iv) Hence deduce that $\tan 2\beta = \frac{2 \tan \beta}{1 - \tan^2 \beta}$. 2**END OF EXAMINATION**

B L A N K P A G E

B L A N K P A G E

The following list of standard integrals may be used:

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

Formb

Half Yearly Ext 1 2011

Q1 a) i) $\cos \frac{7\pi}{6} = -\cos \frac{\pi}{6}$
 $= -\frac{\sqrt{3}}{2}$ ✓

ii) $\tan^{-1}(\sqrt{3}) = \frac{\pi}{3}$ ✓

b) $\sin 4x$ ✓

c) i) $\frac{d}{dx} x^2 \cos x = 2x \cos x - x^2 \sin x$ ✓✓

ii) $\frac{d}{dx} e^{4x} = 4e^{4x}$ ✓

iii) $\frac{d}{dx} (\ln(5x+3)) = \frac{5}{5x+3}$ ✓

d) $\lim_{x \rightarrow 0} \left(\frac{\sin 7x}{x} \right) = 7 \lim_{x \rightarrow 0} \frac{\sin 7x}{7x}$ ✓
 $= 7$

e) i) $\int (1-x)^5 dx = -\frac{(1-x)^6}{6} + C$ ✓

ii) $\int (\cos x - \sin x) dx$
 $= \sin x + \cos x + C$ ✓

iii) $\int \sin 2x = -\frac{1}{2} \cos 2x + C$ ✓
 (or $\sin^2 x + C$
 or $-\cos^2 x + C$)

iv) $\int e^{2-3x} dx = -\frac{e^{2-3x}}{3} + C$ ✓

Q2 a) Amplitude ≈ 3 ✓
 period $T = \frac{2\pi}{2}$ ✓
 $= \pi$ ✓

b) $2x + y - 2 = 0$

$y = -2x + 2$

$m_1 = -2$

$m_2 = 1$

$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$

$= \left| \frac{-2 - 1}{1 - 2 \times 1} \right|$ ✓

$= \left| \frac{-3}{-1} \right|$

$\tan \theta = 3$ ✓

$\theta \approx 72^\circ$ (nearest degree)

c) $\frac{1}{x+5} \geq 1$

$(x+5) \geq (x+5)^2$

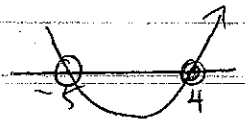
$(x+5)^2 - (x+5) \leq 0$

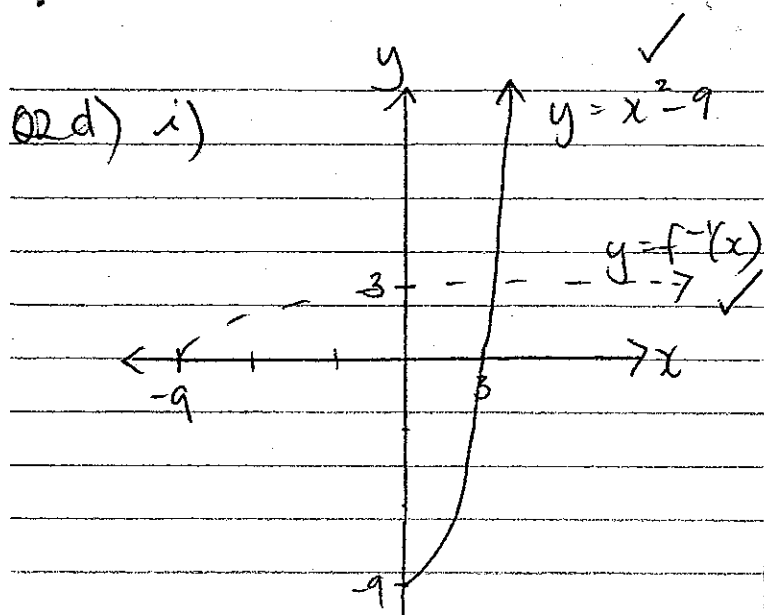
$(x+5)[(x+5) - 1] \leq 0$ ✓

$(x+5)(x+4) \leq 0$ ✓

$-5 < x \leq -4$ ✓

$(x \neq -5)$ ✓





ii) $y = x^2 - 9$
 Swap variables
 $x = y^2 - 9$
 $y^2 = x + 9$
 $y = \sqrt{x + 9}$ Range $y \geq 0$

iii) Domain $x \geq -9$

iv) on same diagram as i)

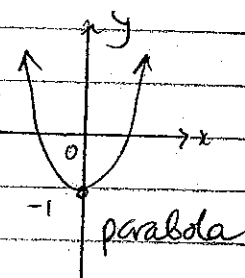
Q3 a) $A(12, -10)$ $B(-3, -5)$
 $2:3$
 $k:l$
 $x = \frac{lx_1 + ky_1}{k+l}$ $y = \frac{ly_1 + ky_2}{k+l}$
 $= \frac{3 \times 12 + 2 \times -3}{5}$ $= \frac{3 \times -10 + 2 \times -5}{5}$
 $= \frac{36 - 6}{5}$ $= \frac{-30 - 10}{5}$
 $= 6$ $= -8$

$P(x, y) = (6, -8)$

b) i) $x = 4t$ $y = t^2 - 1$
 $t = \frac{x}{4}$

$y = \left(\frac{x}{4}\right)^2 - 1$

$y = \frac{x^2}{16} - 1$
 or $x^2 = 16(y + 1)$



ii) Vertex $(0, -1)$

iii) $4a = 16$
 $a = 4$ Focus $(0, 3)$

c) $\cos 2\theta - \sin \theta = 0$ $0 \leq \theta \leq 2\pi$
 $1 - 2\sin^2 \theta - \sin \theta = 0$

$2\sin^2 \theta + \sin \theta - 1 = 0$

$(2\sin \theta - 1)(\sin \theta + 1) = 0$

$\sin \theta = \frac{1}{2}$ or $\sin \theta = -1$

$\theta = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2}$

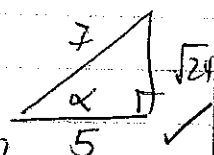
d) $\tan(\cos^{-1}(-\frac{5}{7}))$

let $\alpha = \cos^{-1}(-\frac{5}{7})$

$\cos \alpha = -\frac{5}{7}$

$2\pi \leq \alpha < \pi$ α in 2nd quad.

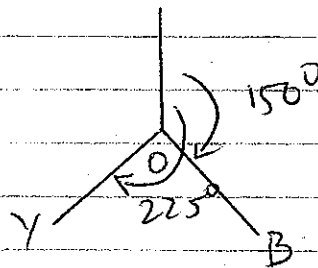
$\tan \alpha = -\frac{\sqrt{24}}{5}$



$$\begin{aligned}
 \text{Q4a) LHS} &= \frac{1}{1-\cos\theta} + \frac{1}{1+\cos\theta} \\
 &= \frac{(1+\cos\theta) + (1-\cos\theta)}{(1-\cos\theta)(1+\cos\theta)} \checkmark \\
 &= \frac{2}{1-\cos^2\theta} \\
 &= \frac{2}{\sin^2\theta} \checkmark \\
 &= 2\operatorname{cosec}^2\theta \\
 &= \text{RHS}
 \end{aligned}$$

$$\begin{aligned}
 \text{d) i) } \angle PYO &= 26^\circ \text{ (alternate } \angle \text{)} \\
 \angle PBO &= 150^\circ \text{ (on } \parallel \text{ lines)}
 \end{aligned}$$

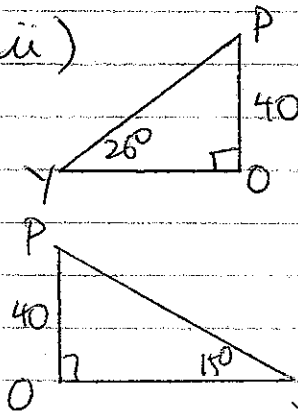
Plan view



$$\begin{aligned}
 \angle YOB &= \text{Bearing of Y} - \text{Bearing of B} \\
 &= 225^\circ - 150^\circ \checkmark \\
 &= 75^\circ \text{ (adjacent } \angle \text{)}
 \end{aligned}$$

$$\begin{aligned}
 \text{b) } \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \tan^2 x \, dx \\
 &= \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} (\sec^2 x - 1) \, dx \\
 &= \left[\tan x - x \right]_{\frac{\pi}{4}}^{\frac{\pi}{3}} \checkmark \\
 &= \left(\tan \frac{\pi}{3} - \frac{\pi}{3} \right) - \left(\tan \frac{\pi}{4} - \frac{\pi}{4} \right) \\
 &= \sqrt{3} - \frac{4\pi}{12} - 1 + \frac{3\pi}{12} \\
 &= \sqrt{3} - 1 - \frac{\pi}{12} \checkmark
 \end{aligned}$$

ii)



$$\tan 26^\circ = \frac{40}{YO}$$

$$\begin{aligned}
 YO &= \frac{40}{\tan 26^\circ} \\
 &= 40 \cot 26^\circ \\
 &\text{or } 40 \tan 64^\circ
 \end{aligned}$$

$$\tan 15^\circ = \frac{40}{BO}$$

$$\begin{aligned}
 BO &= \frac{40}{\tan 15^\circ} \\
 &= 40 \cot 15^\circ \\
 &\text{or } 40 \tan 75^\circ
 \end{aligned}$$

c) let l be the side length

$$\frac{dl}{dt} = 3 \quad V = l^3$$

$$\begin{aligned}
 \frac{dV}{dt} &= \frac{dV}{dl} \times \frac{dl}{dt} \checkmark \\
 &= 3l^2 \times 3 \checkmark \\
 &= 9l^2
 \end{aligned}$$

$$\begin{aligned}
 \frac{dV}{dt} \text{ (at } l=5) &= 9 \times 25 \checkmark \\
 &= 225 \text{ mm}^3/\text{s}
 \end{aligned}$$

In $\triangle YOB$ cosine rule

$$\begin{aligned}
 YB^2 &= BO^2 + YO^2 - 2 \times BO \times YO \cos 75^\circ \\
 &= 40^2 (\tan^2 75^\circ + \tan^2 64^\circ - 2 \times \tan 75^\circ \tan 64^\circ \cos 75^\circ)
 \end{aligned}$$

$$\approx 22674$$

$$YB = 151 \text{ m}$$

Q5 i) Equation of P

$$y = px - ap^2 \quad (1)$$

Equation of Q

$$y = qx + aq^2 \quad (2) \quad \checkmark$$

$$(1) = (2)$$

$$px - ap^2 = qx - aq^2$$

$$px - qx = ap^2 - aq^2$$

$$x(p-q) = a(p-q)(p+q) \quad \checkmark$$

$$x = a(p+q) \quad (3)$$

Sub (3) into (1)

$$y = pa(p+q) - ap^2 \quad \checkmark$$

$$= ap^2 + apq - ap^2$$

$$= apq \quad \checkmark$$

~~Q5 ii)~~ $pq = -2$ $\checkmark$

$y = -2a$ is the locus of T

as a is constant.

b) i) $\sqrt{3} \cos x - \sin x = R \cos(x+\alpha)$

$$\sqrt{3} \cos x - \sin x = R \cos x \cos \alpha$$

$$- R \sin x \sin \alpha$$

$$= R \cos \alpha \cos x - R \sin \alpha \sin x$$

$$R \cos \alpha = \sqrt{3}$$

$$-R \sin \alpha = -1$$

$$R^2 = 4$$

$$R = 2 \quad R > 0 \quad \checkmark$$

$$2 \cos \alpha = \sqrt{3}$$

$$-2 \sin \alpha = -1$$

$$\cos \alpha = \frac{\sqrt{3}}{2}$$

$$\sin \alpha = \frac{1}{2}$$

$$\alpha = \frac{\pi}{6}$$

So $\sqrt{3} \cos x - \sin x = 2 \cos(x + \frac{\pi}{6})$

ii) $2 \cos(x + \frac{\pi}{6}) = 1 \quad -\pi \leq x \leq \pi$

let $u = x + \frac{\pi}{6} \quad -\frac{5\pi}{6} \leq u \leq \frac{7\pi}{6}$

$$2 \cos u = 1$$

$$\cos u = \frac{1}{2}$$

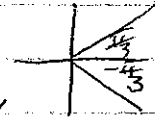
$$u = \frac{\pi}{3} \quad \text{or} \quad u = -\frac{\pi}{3} \quad \checkmark$$

$$x + \frac{\pi}{6} = \frac{\pi}{3}$$

$$x = \frac{\pi}{6}$$

$$x + \frac{\pi}{6} = \frac{\pi}{3}$$

$$x = -\frac{\pi}{3} \quad \checkmark$$



c) A. For $n=0$

$$7^{2n+1} + 11^{2n+1} = 7 + 11 = 18 \text{ div by } 3 \quad \checkmark$$

B. Suppose k is such that $7^{2k+1} + 11^{2k+1}$ is div by 3

for some integer value of $k > 0$

ie. $7^{2k+1} + 11^{2k+1} = 3m$ for some integer m . $\checkmark$

C. The statement must be proved for $n=k+1$

$$7^{2k+3} + 11^{2k+3}$$

$$= 7^{2k+1} \times 7^2 + 11^{2k+3}$$

$$= (3m - 11^{2k+1}) 7^2 + 11^{2k+3} \quad \checkmark$$

$$= 3m \times 7^2 + 11^{2k+1} (11^2 - 7^2) \quad \text{using the inductive hypothesis} \quad \checkmark$$

$$= 3m \times 7^2 + 11^{2k+1} \times 7^2$$

$$= 3(7^2 m + 11^{2k+1})$$

which is divisible by 3 $\checkmark$

So by steps A, B and mathematical induction, the statement is true for $n = 0, 1, 2, 3, \dots$

Ob a) $y = 2 + 3 \sin x$
 $y^2 = (2 + 3 \sin x)^2$
 $= 4 + 12 \sin x + 9 \sin^2 x$

$$V = \pi \int_0^{\frac{\pi}{6}} (4 + 12 \sin x + 9 \sin^2 x) dx$$

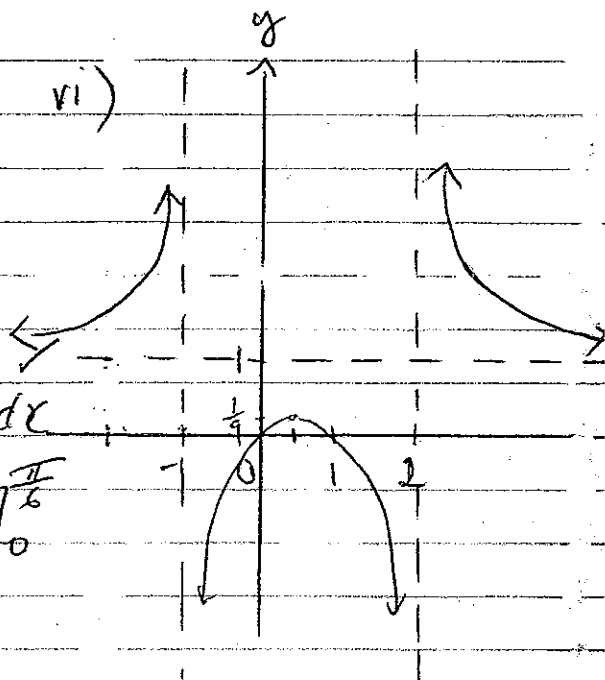
$$= \pi \left[4x - 12 \cos x \right]_0^{\frac{\pi}{6}} + 9\pi \int_0^{\frac{\pi}{6}} \left(\frac{1}{2} - \frac{1}{2} \cos 2x \right) dx$$

$$= \pi \left(\frac{2\pi}{3} - 12 \times \frac{\sqrt{3}}{2} + 12 \right) + 9\pi \left[\frac{x}{2} - \frac{1}{4} \sin 2x \right]_0^{\frac{\pi}{6}}$$

$$= \pi \left(\frac{2\pi}{3} - 6\sqrt{3} + 12 \right) + 9\pi \left(\frac{\pi}{12} - \frac{1}{4} \times \frac{\sqrt{3}}{2} \right)$$

$$= \pi \left(\frac{2\pi}{3} - 6\sqrt{3} + 12 + \frac{3\pi}{4} - \frac{9\sqrt{3}}{8} \right)$$

$$= \pi \left(\frac{17\pi}{12} - \frac{57\sqrt{3}}{8} + 12 \right) u^3$$



b) i) $(0, 0), (1, 0)$ ✓
 ii) vertical asymptotes
 at $x = -1$ and $x = 2$ ✓

iii) $y = 1$ ✓

iv) $f(x) = \frac{x(x-1)}{(x+1)(x-2)}$ ✓

$$f'(x) = \frac{(x+1)(x-2)(2x-1) - x(x-1)(2x-1)}{(x+1)^2(x-2)^2}$$

$$= \frac{(2x-1)[x^2 - x - 2 - (x^2 - x)]}{(x+1)^2(x-2)^2}$$

$$= \frac{-2(2x-1)}{(x+1)^2(x-2)^2} \checkmark$$

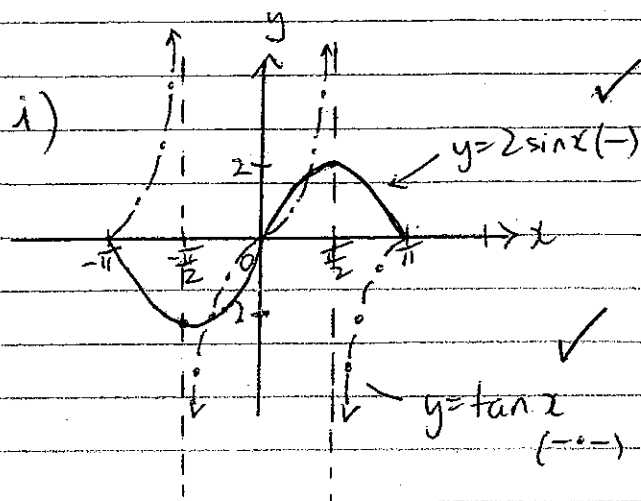
v) $f'(x) = 0$ when $x = \frac{1}{2}$
 $y = \frac{1}{9}$ ✓

x	0	$\frac{1}{2}$	1
$f'(x)$	+	0	-

$\left(\frac{1}{2}, \frac{1}{9} \right)$ is a maximum turning point. ✓

Q7

a) i)



Given the x-coordinate of B is 1
 iii) + let x-coordinate of C be β
 $(\alpha + \beta = -\frac{b}{a})$

$$1 + \beta = \frac{2m^2}{1+m^2}$$

$$\beta = \frac{2m^2}{1+m^2} - 1$$

$$= \frac{2m^2 - 1(1+m^2)}{1+m^2}$$

$$= \frac{m^2 - 1}{m^2 + 1}$$

$$ii) 2 \sin x = \tan x \quad -\frac{\pi}{2} < x < \frac{\pi}{2}$$

$$2 \sin x = \frac{\sin x}{\cos x}$$

$$2 \sin x \cos x - \sin x = 0$$

$$\sin x (2 \cos x - 1) = 0$$

$$\sin x = 0$$

$$\cos x = \frac{1}{2}$$

$$x = 0$$

$$\text{or } x = \pm \frac{\pi}{3}$$

y-coordinate of C is given by
 $y = m \left(\frac{m^2 - 1}{m^2 + 1} - 1 \right)$

$$= m \left(\frac{m^2 - 1 - m^2 - 1}{m^2 + 1} \right)$$

$$= \frac{-2m}{m^2 + 1}$$

$$iii) 0 < x < \frac{\pi}{3} \quad \text{or} \quad \frac{\pi}{2} < x < \frac{\pi}{3} \quad \checkmark$$

$$b) i) B(1,0) \quad y - 0 = m(x - 1) \quad \checkmark$$

$$y = mx - m$$

ii) B & C are the solutions of the simultaneous equations

$$y = mx - m \quad (1)$$

$$x^2 + y^2 = 1 \quad (2) \quad \checkmark$$

Sub (1) into (2)

$$x^2 + (mx - m)^2 = 1 \quad \checkmark$$

$$x^2 + m^2 x^2 - 2m^2 x + m^2 = 1$$

$$x^2 (1 + m^2) - 2m^2 x + (m^2 - 1) = 0$$

as required.

$$C \left(\frac{m^2 - 1}{m^2 + 1}, \frac{-2m}{m^2 + 1} \right)$$

iv) $\triangle OBC$ is isosceles (radii)

$$\alpha = 2\beta \quad (\text{exterior angle of } \triangle)$$

$$\tan \alpha = \tan 2\beta \quad (\alpha \text{ is acute})$$

$$\text{But } \tan(\pi - \beta) = m$$

$$\text{so } \tan \beta = -m \quad \checkmark$$

$$\tan(\pi - \alpha) = \text{gradient of } \alpha$$

$$-\tan \alpha = \frac{-2m}{1+m^2} \div \frac{m^2 - 1}{m^2 + 1}$$

$$= \frac{-2m}{m^2 - 1}$$

$$\text{So } \tan 2\beta = \frac{2m}{m^2 - 1}$$

$$= \frac{-2 \tan \beta}{\tan^2 \beta - 1} \quad \checkmark$$

$$= \frac{2 \tan \beta}{1 - \tan^2 \beta}$$

as required.